

AI Operations (AIOps) — DA3408

Assignment 1

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Question 1

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1. Entanglement (CACE) - Changing the rounding logic of one input feature silently shifted the model's behaviour on an apparently unrelated feature.
2. Undeclared Consumer - Another team is using the model's output without telling anyone, so the ML team cannot change it safely.
3. Configuration and Glue-Code Debt - The pipeline is a pile of scripts with nothing to run them in order, and settings are spread all over them.

2

Lets try to resolve (b)

1. **The model should be properly released.** The trained model is registered under a name and then moved through the stages like Development → Staging → Production. Marketing team should be allowed to only use Production model and nothing else. The ML team can keep training and testing other models in development stage.
2. **Restrict the user for each model.** Have a consumers.yaml where it is listed which team is using the model, and what they are using it for. The same thing is also added as a tag on the MLflow run, so it shows up on the run page too.

Question 2

Best run: `mlp-lr0.001-bs24` leads on every measure, with peak validation accuracy 0.9752, test accuracy 0.9728, macro F1 0.9725, and the lowest final training loss 0.0147. Similar to it `mlp-lr0.001-bs208` matches on validation but trails by 0.0035 on test accuracy, so the smaller batch wins.

Overfitting: that same run diverges most clearly. Training loss falls to 0.0147, the lowest of the six, yet validation accuracy peaks at 0.9752 and settles to 0.9721. Across the grid, validation flattens after epoch 10 while training loss keeps descending. `mlp-lr0.01-bs24` stopped early at epoch 17 with an oscillating loss, indicating instability rather than overfitting.

Larger effect: learning rate. At batch size 24, test accuracy moves 0.9682, 0.9728, 0.9565 across the three rates, a spread of 1.6 points, with 0.01 unstable and 0.0002 underfitting. Batch size at a fixed learning rate shifts accuracy by only about 0.35 points.